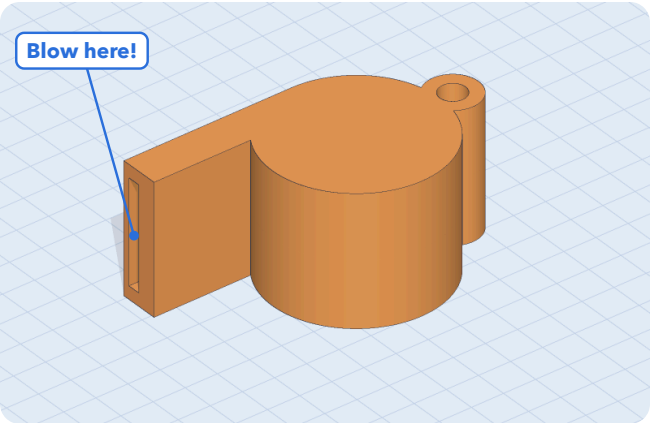
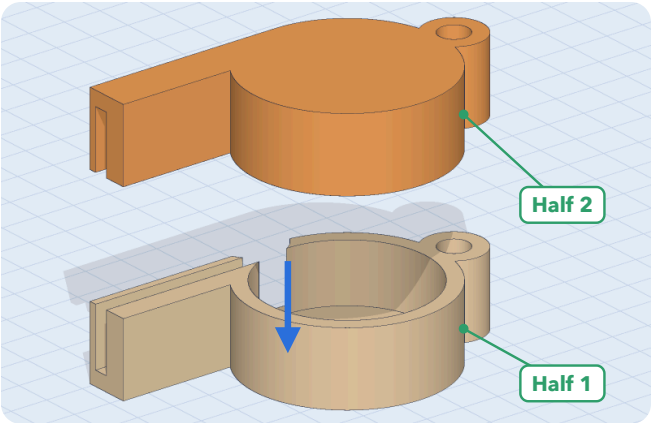
Project 2 | TinkerCad | About 45 min

Make a 3D Printed Whistle in TinkerCad

Design a whistle that really works, and blow it for real. It prints in one piece!



This is what you're making. Blow into the thin slot.



Here's the trick: you build one half, then copy it, flip it over and stack it.

How does a whistle work?

You blow air into a **thin tunnel**. The air shoots out and hits a **sharp edge** at the sound window. The air splits and makes the air inside the round **chamber** spin and shake really fast. Fast shaking air = **sound!**

Here's the plan

You'll build just the **bottom half** first: a **round barrel** plus a **mouthpiece**, scooped out with **hole** shapes. With the top off, you can see exactly what you're making! Then you **mirror** it to make the top half and join them.

Hole

A see-through shape. When you group it with a solid shape, it cuts that space away.

Group

Joins shapes into one. This is when the holes actually get cut.

Align

Lines up the edges or centers of shapes so they fit together perfectly.

NAME YOUR DESIGN!

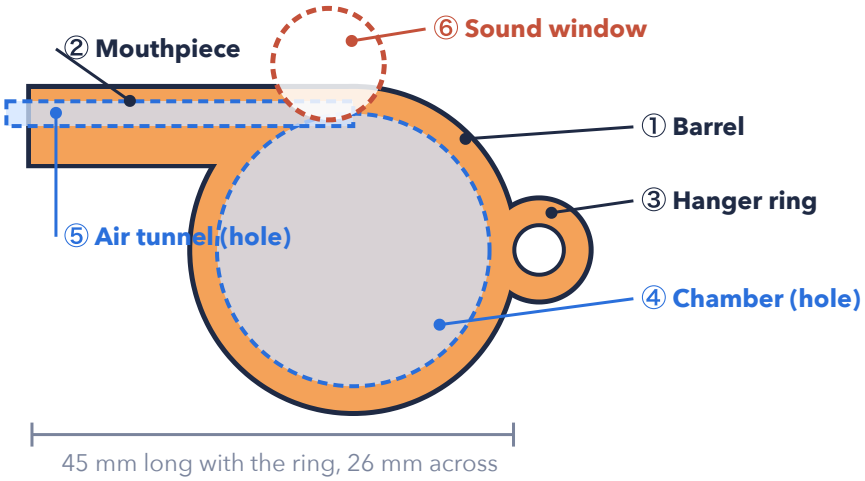
Your teacher downloads everyone's whistles from TinkerCad to print them. Name yours **Whistle - First Name Last Initial** (like **Whistle - Maya R**) so we know it's yours. You'll do this in Step 1.

GOLDEN RULE

Made a mistake? Press `Ctrl` + `Z` to undo. You can't break anything. 😄

🗺️ The Parts Map

This is your half whistle seen from the **top**. Solid parts are orange. **Holes** are blue and dashed (the sound window is red so you can spot it, and it pokes out past the back on purpose).



	PART	SHAPE	SOLID OR HOLE	SIZE (MM)	HEIGHT OFF GROUND
①	**Barrel**	Cylinder	Solid	`26 × 26 × 10`	`0`
②	**Mouthpiece**	Box	Solid	`26 × 6 × 10`	`0`
③	**Hanger ring**	Cylinder	Solid	`8 × 8 × 10`	`0`
	...ring hole	Cylinder	**Hole**	`4 × 4 × 12`	`0`
④	**Chamber**	Cylinder	**Hole**	`22 × 22 × 12`	`2`
⑤	**Air tunnel**	Box	**Hole**	`28 × 2 × 12`	`2`
⑥	**Sound window**	Cylinder	**Hole**	`9 × 9 × 12`	`2`

HOW TO READ THE SIZES

Sizes go **long × wide × tall**. Looking from the top, "long" is left-right on your screen and "wide" is up-down. "Height off ground" is how high you lift the shape with the black cone. Keep this page next to you!

HOW TO READ THE PICTURES

In the next pages, the **newest part is bright** and older parts are pale. **Holes are blue** here, but gray and stripy in TinkerCad. They stick up past the top of your half on purpose, so they cut all the way through. The colors on your screen will be different too, and that's fine.

1 Get set up

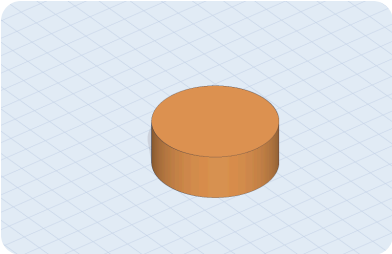
- [ ] Open TinkerCad and click **Create → 3D Design**.
- [ ] **Name it right away!** Click the silly random name at the top left, type **Whistle – First Name Last Initial**, and press `Enter`.
- [ ] Check that **Snap Grid** (bottom right) says **`1.0 mm`**. Then click **Top** on the view cube so you're looking straight down.

Two tricks you'll use a lot

Typing a size: click a shape, hover over a white corner square, click a number that pops up, type the new number and press `Enter`. The **height** number is on the white square on top.
Lifting a shape: grab the **black cone** on top of the shape, click the number and type how high it goes.

2 The barrel ①

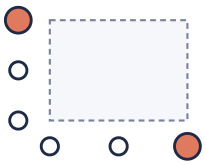
- [ ] Drag a **Cylinder** onto the middle of the workplane.
- [ ] Make it **`26 × 26 × 10`**. That's half as tall as the finished whistle.



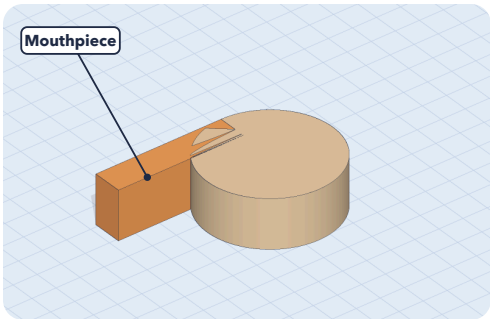
You should see: a short, fat cylinder.

3 The mouthpiece ②

- [ ] Drag a **Box** onto the workplane, to the **left** of the barrel and a little **lower** on your screen.
- [ ] Make it **`26 × 6 × 10`** (26 long, 6 wide, 10 tall).
- [ ] The box is still selected. Hold `Shift` and click the barrel, so both are selected.
- [ ] Click **Align** (or press `L`). Click the **right** dot in the left-right row, and the **top** dot in the up-down row (see the picture).
- [ ] Click the box by itself and press `←` **13 times**.



Click the two orange dots: **right** and **top**



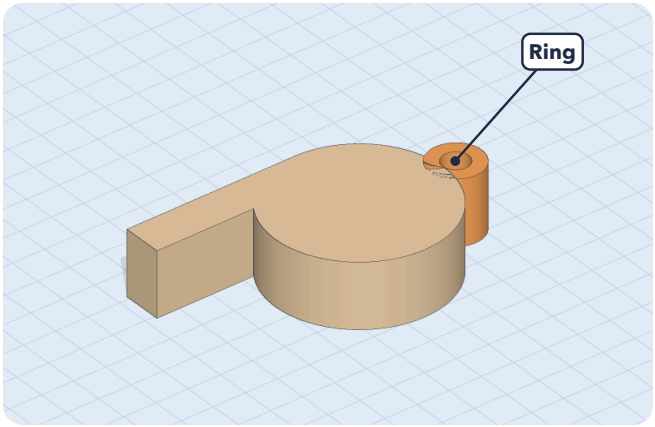
You should see: the backs line up, and the mouthpiece stops at the **middle** of the barrel.

4 The hanger ring ③

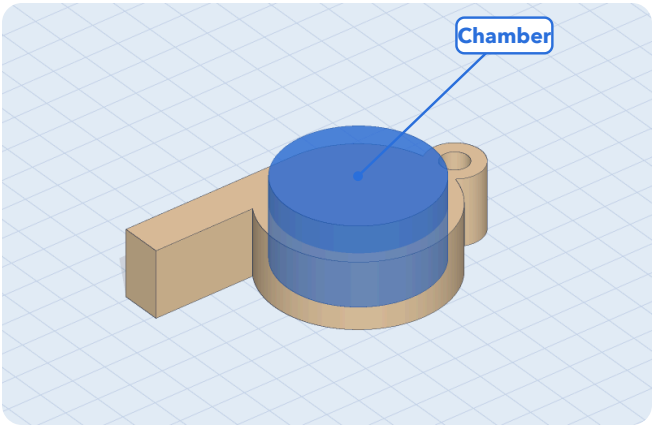
- [ ] Drag a **Cylinder** onto the empty workplane and make it `8 × 8 × 10`.
- [ ] Drag another **Cylinder** onto it, click **Hole**, and make it `4 × 4 × 12`. Press `D` to drop it down to the workplane.
- [ ] Select both, **Align** both **middle** dots, then **Group** (`Ctrl` + `G`). Now it's a ring!
- [ ] Drag the ring onto the barrel and press `D` to drop it down. Shift-click the barrel, **Align** both **middle** dots.
- [ ] Click the ring by itself and press `→` **15 times**. It now sticks out the right side of the barrel.

5 The chamber ④

- [ ] Drag a **Cylinder** onto the barrel. Click **Hole**. Make it `22 × 22 × 12`.
- [ ] Shift-click the barrel, **Align** both **middle** dots.
- [ ] Click the hole by itself and set its height off the ground to `2`. That leaves a floor 2 mm thick. The hole sticks up past the top of the barrel, and that's just what we want!



After Step 4: the ring sticks out on the right, on the side away from the mouthpiece.



After Step 5: the chamber hole is centered in the barrel, with an even border all around.

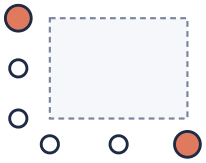
ALIGN TIP

Align should move the **new** shape, not the big one. If the wrong shape moves, press `Ctrl` + `Z`, move the new shape so it sits more on top of the big one, and try again.

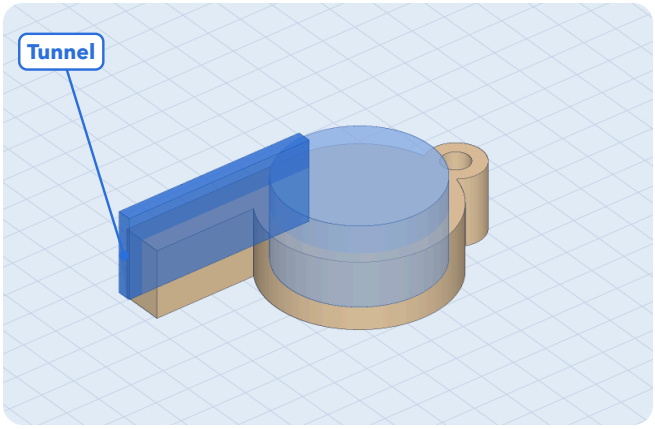
6 The air tunnel ⑤ and sound window ⑥

These two holes are what make the whistle whistle, so be careful here!

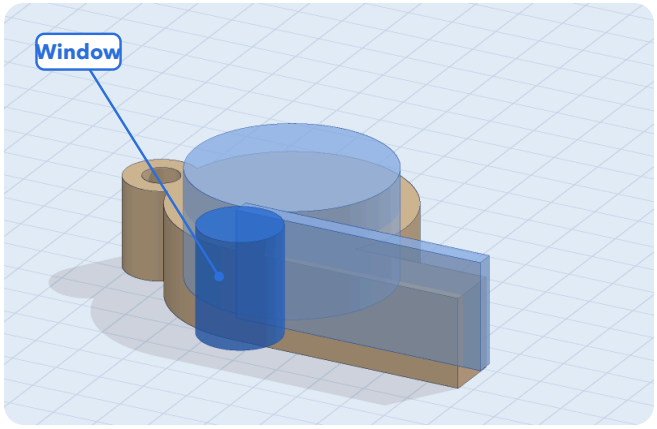
- [ ] **Air tunnel:** drag a **Box** onto the mouthpiece, click **Hole**, and make it `28 × 2 × 12`. Move it so it sticks out past the open (left) end, but **not** past the middle of the barrel.
- [ ] Shift-click the mouthpiece. **Align**: **right** dot and **top** dot.
- [ ] Click the tunnel by itself. Press ↓ **1 time**, then set its height off the ground to `2`.
- [ ] **Sound window:** drag a **Cylinder** onto the barrel, click **Hole**, and make it `9 × 9 × 12`.
- [ ] Shift-click the barrel. **Align** both **middle** dots, just like the chamber.
- [ ] Click the new hole by itself. Press ← **2 times** and ↑ **15 times**, then set its height off the ground to `2`.



Air tunnel dots: **right** and **top**



Air tunnel: a thin slab that pokes out of the end of the mouthpiece.



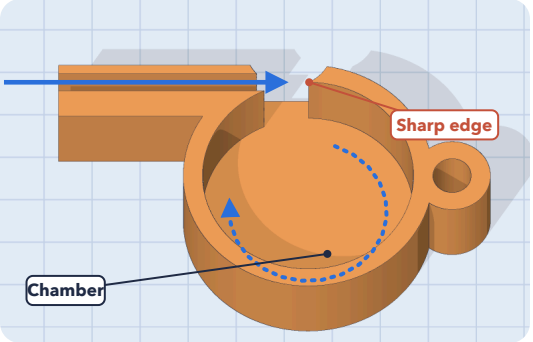
Sound window: turned around to the back. A round hole that pokes out of the back wall, right next to the barrel.

COUNT CAREFULLY!

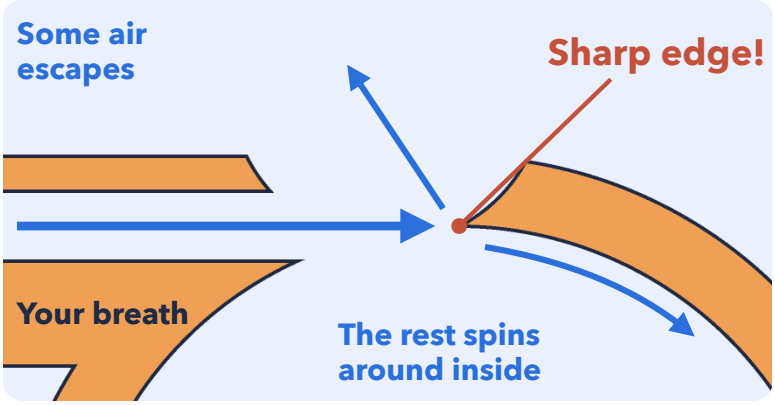
The curve of this round hole is what carves the **sharp edge** that makes the sound. If it's even a little off, the edge misses the air and you won't get a whistle. Check your presses: ← **2** and ↑ **15**. You'll see the edge on the next page.

7 Group your half and look inside

- ☐ Press `Ctrl` + `A` to select everything.
- ☐ Click **Group** (or `Ctrl` + `G`). The holes disappear and get cut out!
- ☐ Click **Top** on the view cube, then drag the view cube to tilt it a little, so you can look down inside.



You should see: your half with its top off: a thin tunnel, the round chamber, and the window cut through the back wall where they meet.



Close-up from the top: the wall comes to a **thin point**, right in the middle of the air stream. Find it on your screen!

How your whistle works

1. Your breath goes along the thin tunnel, like the straight arrow.

2. It hits the **sharp edge** and splits. Some escapes out of the window. The edge has to be sharp and thin: a thick, flat wall would just block the air, like blowing against your hand.

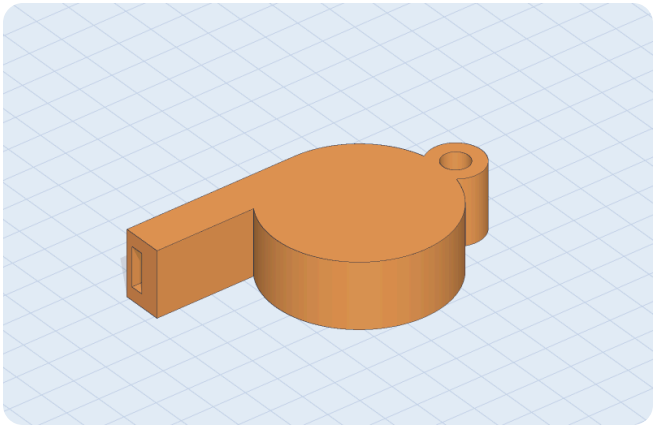
3. The rest spins around the chamber, like the dotted arrow, and shakes the air. That's the sound!

UH OH?

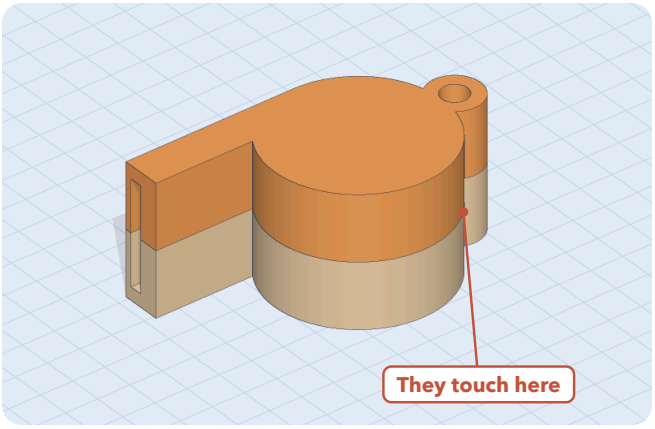
Doesn't match the picture? Press `Ctrl` + `Z` to undo the group, then check the hole sizes, heights and arrow presses against the Parts Map. Fix it now, because the top half will be a copy of this one!

8 Mirror it to make the top

- [ ] Click the view cube's **Home** button (the little house) so you can see your half from the side. Click your half to select it.
- [ ] **Duplicate:** press `Ctrl` + `D`. It looks like nothing happened, but now there's a copy sitting exactly on top of your half, and the copy is selected. **Don't click anywhere else!**
- [ ] **Mirror:** press `M`. Arrows appear around the shape. Click the arrow that flips it **upside down**. Wrong arrow? Press `Ctrl` + `Z` and try another one.
- [ ] **Stack:** grab the **black cone** on top and set its height off the ground to `10`. Now it sits right on top of Half 1.
- [ ] **Join:** press `Ctrl` + `A`, then `Ctrl` + `G` to group the two halves into one whistle.



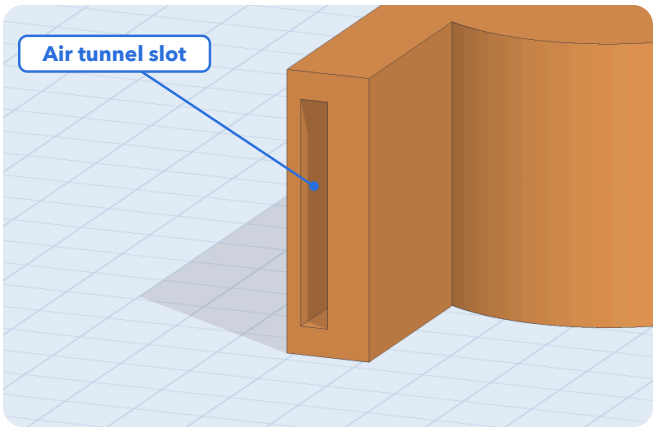
After Mirror: the top looks flat and closed. The two halves share one spot for now.



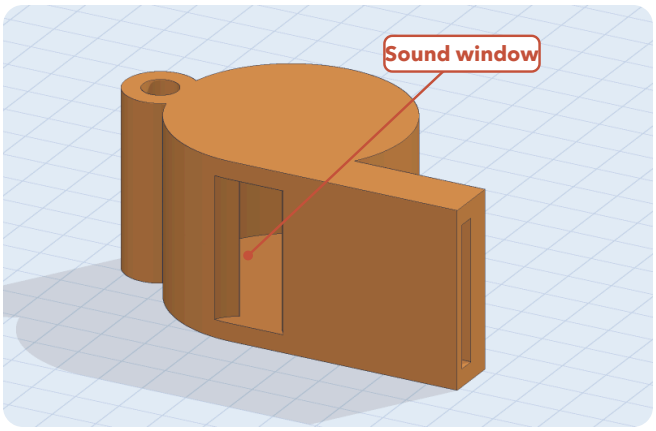
After Stack: one half on top of the other, twice as tall. (Colors are just to show the halves.)

Find the two openings

Drag the view cube to turn your finished whistle around:



The end of the mouthpiece: a thin slot. This is where you blow.



The back: a rounded opening right next to the barrel.

9 Final check ✅

- ☐ It's one piece, and it looks like the Parts Map from the top: barrel, mouthpiece and hanger ring.
- ☐ There's a thin slot at the end of the mouthpiece and a rounded window on the back, next to the barrel.
- ☐ It's about `45 × 26 × 20` mm and sitting flat on the workplane.

10 Save it and hand it in 💾

- ☐ Check the name at the top left says **Whistle – First Name Last Initial**. Not right? Click it and fix it.
- ☐ TinkerCad saves by itself while you work. To be sure, click the **TinkerCad logo** (top left) to go back to your designs. Your whistle should be there with **your name on it**.
- ☐ Tell your teacher your whistle is ready! Your teacher will download it from TinkerCad and 3D print it for you. 🖨️
- ☐ **Done early?** Make it yours! Add your initials on the side of the barrel with the **Text** shape (make it stick out just 1 mm), or try a different color. Don't change the holes, or it might stop whistling!

🤔 Think about it

1. Which part of the whistle makes the sound? Explain it in your own words.

2. If you made another whistle, what would you change?

CREDITS

This guide is adapted from "3D Printing Whistle" by **Ally Zhao** on Instructables, used under a Creative Commons BY-NC-SA license. Thank you, Ally! **Changes:** the original was made in 123D Design; this version rebuilds the whistle (barrel, mouthpiece, air tunnel, sound window and hanger) with TinkerCad shapes, using the original's sizes and its idea of building one half and mirroring it, and is rewritten as a step-by-step classroom worksheet with new pictures. To keep it simple, it leaves out the original's pea, makes the original's tapered edge at the sound window with one round hole, and runs the hanger the full height of the whistle. This adapted version is shared under the same license, CC BY-NC-SA 4.0: free to share and change for non-commercial use, as long as you give credit and share alike. TinkerCad is a trademark of Autodesk; this guide is not made by or affiliated with Autodesk or Instructables.